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REACT Initiative: Army Aviation's Process to Winning the Drone Wars

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Executive Summary

This paper introduces the Real-time Enemy Assessment and Command Tool (REACT) initiative as a holistic solution to enable tactical commanders to dominate the drone battlespace. By integrating material, personnel, leadership, education and training aspects of doctrine, organization, training, materiel, leadership, education, personnel, facilities, and policy (DOTMLPF-P), REACT positions Army Aviation at the forefront of drone warfare. REACT enables the commander to direct their human organization's efforts more efficiently by reducing the staff workload fighting within the drone domain. The REACT initiative equips ground tactical commanders with a semi-automated military decision making process (MDMP) for timely and efficient drone operations, necessary personnel structures, and a training framework to build a force proficient in AI-enhanced combined arms warfare.

The Human Dilemma in a Drone War

In Iraq, my unit was repeatedly targeted by one-way drones, not only designed to kill but also to erode morale. The force fighting us was a poorly equipped, under funded, and a disorganized enemy. During my seven month deployment, the drone strikes killed three people and injured many more. We were lucky. An organized enemy, synchronized across domains, could have decimated the Aviation Task Force. This experience led me to a simple realization: the most effective way to counter drones is with our own. Army senior leaders understand that drones are today's warfare's greatest threat and will be even more prevalent in future battles. The clear gap that Army Aviation faces is how to synchronize friendly drones to defend critical assets and people adeptly. Efficiently implementing drones will be vital to winning a drone war, especially against an enemy that has a numerical advantage in soldiers and material. The REACT initiative provides the tools, personnel, leadership, education and training commanders need to implement MDMP given the most challenging conditions and to synchronize drone assets to win the drone wars.

REACT Program: A Semi-Automated MDMP Tool

The REACT program will be a desktop application that automates MDMP. The program will have user-defined inputs, access to live sensor data, and access to pre-loaded databases. Servers will provide continuous program updates while integrated sensors from the battlefield will provide the most updated data available to create a full picture of the battlefield. The program integrates with existing C2 architecture and functions as a decision-support tool that uses a feedback loop to continuously improve its algorithm. A high level depiction of the feedback loop demonstrated by Figure 1 shows how REACT will integrate across time and space.

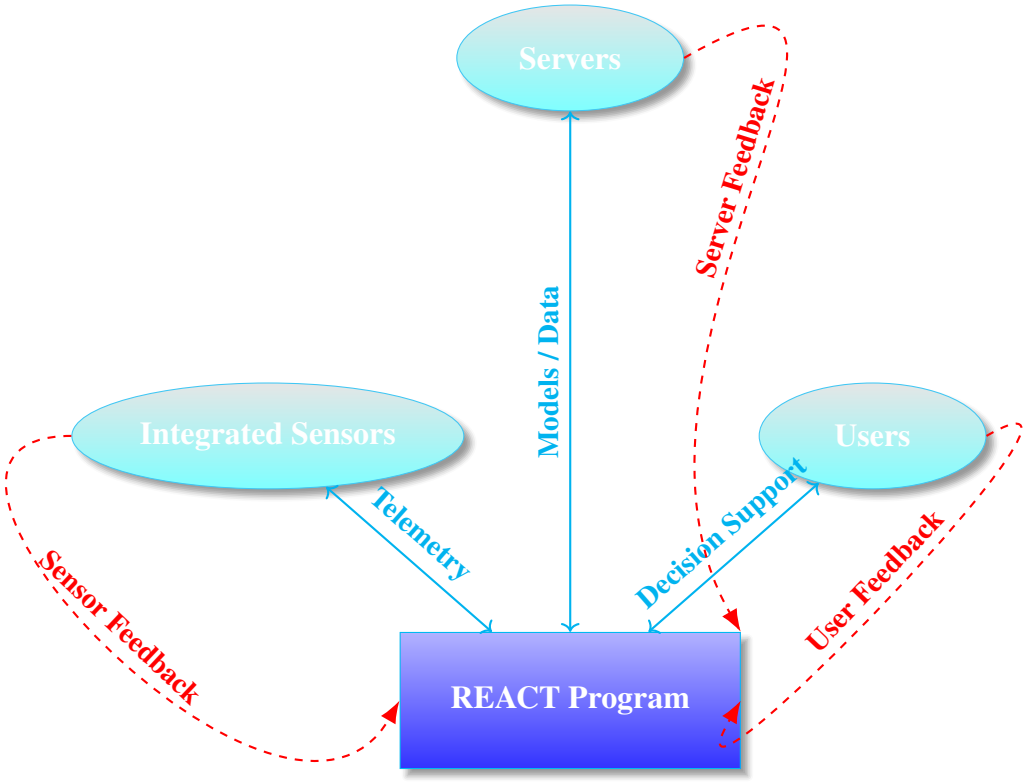


Figure 1: REACT Feedback Loop: REACT exchanges data, telemetry, and decision support with external systems. Feedback from each component drives adaptive learning. Servers would incorporate Anduril's Lattice Mesh and Lockheed Martin's FAAD technology. Integrated sensors would include radars, UAS, etc. The REACT program would be an application assessable from the Menace platform.

The Army's fragmented command and control ecosystem presents a critical vulnerability. Recently, the Army contracted Anduril to build the next-gen C2 ecosystem.[1] Anduril proposed a solution called "Lattice Mesh" to the current silo confined systems within each warfighting function. Lattice Mesh "breaks down traditional data silos by integrating and normalizing heterogeneous data in real-time serving as a translator that can speak the unique language of each service, platform, robot, etc."[2] The ability to "speak the unique language of each robot" will be instrumental in REACT's design. The Army will continue to purchase commercial off the shelf (COTS)

where vendors will likely produce drones and robots using different software languages. Lattice Mesh will be the bridge between these software languages and allow commanders a centralized node to command and control the plethora of assets he or she will possess. The REACT system should incorporate the advantages of the Lattice Mesh architecture as well as Anduril's *Menace* Command, Control, Communications, and Computing (C4) solutions to outpace adversaries drone and counter-drone capabilities. Air defense artillery (ADA) battalions already use another critical capability that could aid combat aviation brigades (CABs) called the FAAD system designed by Lockheed Martin. REACT would be designed to integrate FAAD software and AI into its decision making process.

CABs across the Army are still far behind Secretary Hegeseth's memorandum stating the expectation that "next year [Secretary Hegeseth] expect to see [weaponizing and training of small drones] integrated into all relevant combat training, including force-on-force drone wars."[3] The only drones CABs are prepared to fight with today are MQ-1C Gray Eagles. CABs must invest quickly in the procurement and training of small drones, counter-drone munitions, C2 architecture, and robotics and fully integrate these assets in order for REACT's integration. Without these enabling assets, CABs are likely to fare no better than the Russian strategic bombers after June 1st, 2025.[4]

REACT's Mission Analysis Process

REACT uses artificial intelligence, GPS data, and structured databases to analyze the PMESII-PT variables (political, military, economic, social, information, infrastructure, physical environment, and time). Once the operational environment is modeled, REACT proceeds to generate and simulate potential courses of action (COAs), leveraging both historical precedent and real-time data. This approach enables faster and more accurate MDMP completion, as illustrated in Figure 2.

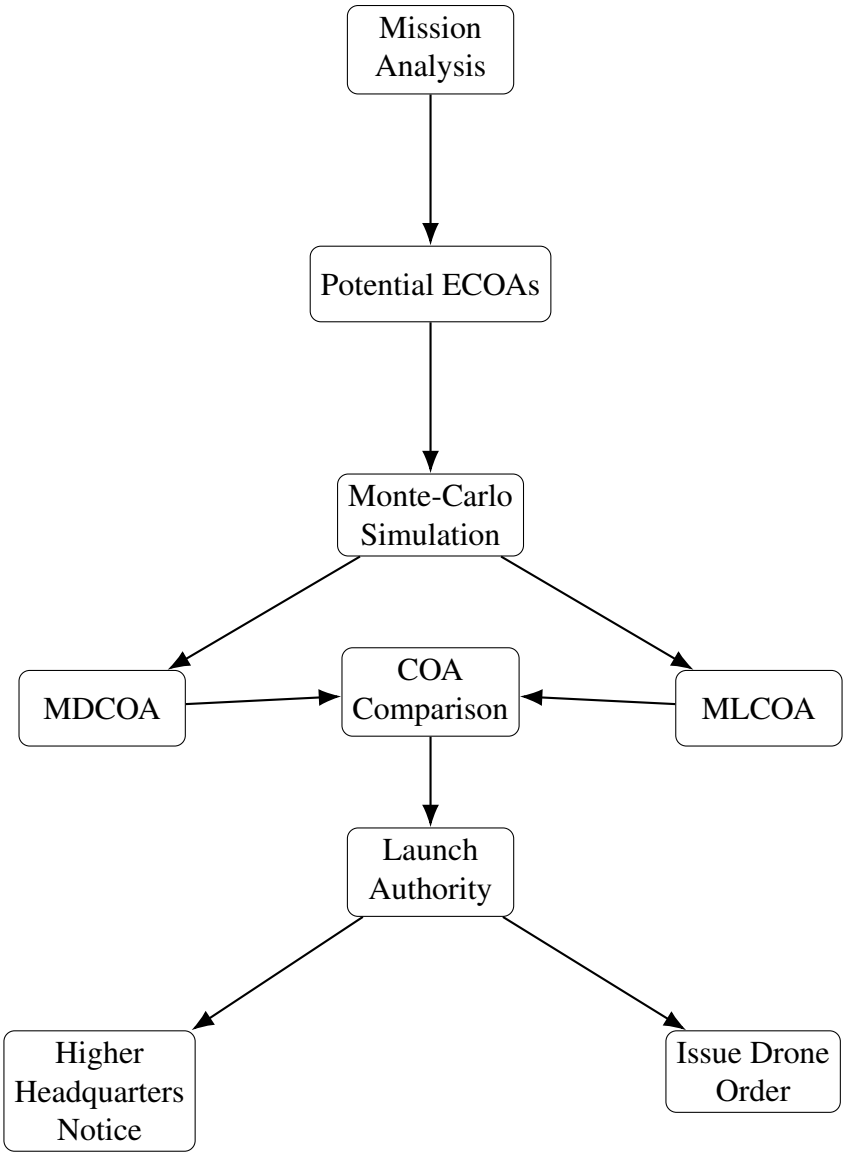


Figure 2: *REACT Program MDMP Steps 3-7.*

Artificial Intelligence Role in REACT

Artificial Intelligence plays a central role in enhancing REACT's utility. Through supervised and reinforcement learning, AI draws insights from previous mission outcomes and after action reports. Alerting features highlight anomalies such as enemy movement or unexpected weather patterns. AI also supports virtual wargaming and streamlines simulation processes via neural networks and qualitative reasoning models.[5] Natural language processing further enables REACT to produce succinct reports for higher echelons automatically.

Monte Carlo Probabilistic COA Comparison

Monte Carlo simulations provide probabilistic insights rather than deterministic conclusions which more accurately incorporates the "art" of war. Although Monte Carlo modeling brings scientific rigor to wargaming, the launch authority must verify and refine the final plan. Operations research already incorporates Monte Carlo simulations into strategic planning cycles. This paper argues the introduction of drones on the battlefield necessitate a rapid decision support tool at the tactical brigade level. While we expect future Soldiers to pick up a first person view (FPV) drone and utilize it like a M4 rifle, not every robot on the battlefield will be quite as expendable. Scarcity of drones and drone resupply will render drones and their operators high-value targets and thus, require careful, concise incorporation into mission planning especially for defensive operations where the tempo is controlled by the adversary.

For each simulation run, REACT will simulate an enemy COA, resource availability, environmental factors, operational constraints, and mission outcomes. Planners must assign each variable a known or modeled distribution and randomize it during simulation. Table 1 lists proposed variables to randomize per run.

Table 1: Variables to Randomize Per Monte Carlo Simulation Run

Variable	**Distribution Example**
Enemy COA Choice	Categorical (weighted: e.g., Flank 40%, Frontal Assault 50%, Ambush 10%)
Enemy Reaction Time	Normal distribution ($\mu = 5$ min, $\sigma = 2$ min)
Drone Failure Rate	Bernoulli distribution (e.g., $p = 0.02$ per sortie)
Ammo Resupply Time	Uniform distribution (15 to 30 minutes)
Logistic Resupply	Beta Distribution (By Supply Class)
Civilian Density	Triangular distribution (Low = 0, Peak = 5, High = 10)
Weather Degradation	Discrete categories (Clear, Light, Heavy with assigned probabilities)
Mission Time Available	Normal distribution ($\mu = 45$ min, $\sigma = 10$ min)
Resource Usage Rates	Normal or empirical (based on unit SOPs or historical data)

A simulation run can consist of tens of thousands to millions of "steps". Within a step,

defined by a finite small interval of time, each asset friendly and enemy will undergo an action or series of actions (shoot, move, communicate). REACT will pixelate the battlefield into appropriately sized grid squares and will simulate through time and space until the commander's intent is achieved or deemed unachievable within the COA. The REACT program will incorporate parallel processing to execute several "chunks" of runs simultaneously. Where each run is a separate COA that is feasible, acceptable, suitable, distinguishable, and complete. The final output will provide a statistical probability and uncertainty of achieving the commander's intent. The output will not only enable commanders to visualize how to use their assets but will synchronize those efforts once the commander makes a decision to pursue a COA.

Feedback Loop: Real-Time Adaptation

While army operations research has pursued models to simulate theater conflicts, campaigns and more complicated simulations, REACT differs starkly by its use of live feedback loops to become a true battlefield tool. The modern transparent fight allows REACT to achieve what no prior simulation could achieve: view current opposing forces and maneuvers via the sensor network and rapidly issue optimized orders that will be followed without hesitation or error, and then learn from the outcome to become more efficient and accurate. "Many models use scripted force allocations as inputs to the simulation. These scripts do not adapt to changes in situation during the course of the simulated battle."[6] REACT knows with reasonable high level of confidence the current enemy composition and disposition to tailor its output, then once the operation is complete, REACT can update those enemy compositions and dispositions based on the sensor feedback. REACT maintains a dynamic tally of enemy movements unmatched by any current model. While REACT could be used to model human on human fights the primary focus on the software would be to win the drone battle. This avoids the common model pitfall that "most models do not adequately reflect the competitive nature of decision-making between opponents on a battlefield" in other words the "art" or "human" aspect of war. With REACT winning the drone battles, commanders can focus on how to effectively fight their human organization.

UAS Brigade Team

In addition to the material required to compete against adversaries and survive in a battlespace swarmed with drones, the CAB lacks sufficient personnel to manage drones at the squad level. Instinctively there are three options:

1. **Retrain and qualify the current formation** that is already largely undermanned by MTOE to operate and maintain drones. Previously, this option was unfeasible; however, the dismantlement of the air cavalry squadrons (ACS) may provide an opportunity to integrate the potential surplus within the Apache community to become Aviation's future sUAS operators and mechanics. This method would still require the disbursement of the ACS Soldiers across the other battalions within the CAB, whose reconnaissance background makes them ideal for tactical drone employment.

2. **Add positions to the MTOE** within the brigade that can manage, operate, synchronize, and maintain the brigade's robot and drone assets. These positions would present a team of Soldiers that are trained and proficient in drone warfare and capable of training others within the formation to operate and maintain drones down to the squad level.

3. **Combine options one and two.** The Soldiers brought over from the ACS would serve as the primary operators and maintainers, while the added positions would function as brigade-level managers and asset synchronization planners.

Regardless of the COA chosen by Army leaders, each brigade must designate at a minimum a launch authority, an airspace manager, a UAS subject matter expert, a UAS senior NCO to advise the launch authority and commander on the utilization of his or her assets, and a UAS senior NCO that can serve as the senior maintainer. This paper proposes an optimized solution for the integration of the REACT program of a six man team organized under BDE HHC, as illustrated in Table 2.

Table 2: REACT Minimum Personnel Roles and Tasks

MOS	Rank	Position	Task
15B	MAJ	Launch Authority & Team Leader	Lead team, assess risk, and operate REACT
140A	CW3	Air and Missile Defense Systems Integrator	Manage airspace deconfliction and UAS coordination
150U	CW3	Tactical Unmanned Aerial Systems Operations Officer	Operate REACT, oversee UAS mission planning, airspace integration, and sensor tasking
15X	SFC	Senior Adviser	Advise on tactical UAS employment and support commander/team leader decision-making
15X	SSG	UAS Senior Operator	Execute launch, recovery, and real-time sensor operations for drone and robot missions
15X	SSG	UAS Senior Maintainer	Maintain system readiness and troubleshoot REACT-integrated UAS platforms

Training AI-Enhanced Combined Arms Officers

Most Soldiers have limited experience with drones and even fewer are competent FPV drone operators. The Aviation Center of Excellence (AvCOE) at Ft Rucker designed a UAS Lethality Course (UALC) scheduled to begin August 4th, 2025. This course trains Soldiers in the "use of lethal UAS munitions, tactics, UAS integration with TTPs at the battalion and higher echelons and [provides] the ability to train operators at home station to increase lethality for the ground force commander."[7] The course program of instruction (POI) covers everything from UAS characteristics to CAD innovation and 3D printing to live flight training with grenade employment. An impressive feat, the course was designed, resourced and implemented in only three months. While operator training has begun, officer training remains underdeveloped. I propose a parallel course designed specifically for officers at aviation captain's career course (AVCCC). The course content would be added at the end of AVCCC's POI and would take approximately three additional weeks. The course's intent is to train the world's most capable UAS operators, proficient in tactical employment, system mastery, AI integration, and multi-domain synchronization. These operators will

support brigade combat teams and REACT-driven MDMP operations through decisive UAS employment in denied and contested environments. The course would consist of three phases outlined below:

Phase I: Foundational Training

- **Technical Certification:** Drone maintenance management, Multi-platform UAS flight, payload configuration, Tactical employment of drones and robotics.
- **REACT Familiarization:** Basic input integration, variable entry, and reading AI-generated ECOA outputs.

Phase II: Classroom Practical Exercises and Simulations

- **Virtual Integration:** Simulated UAS missions in electronic warfare contested, multi-domain environments.
- **Joint Integration:** Coordinate with artillery, aviation, infantry for ISR-to-strike loops and hunter-killer tactics.

Phase III: Field Exercise

- **Combat Judgment:** Use of tactical battle scenarios and rapid MDMP execution using REACT ECOA recommendations.
- **Autonomous Planning:** Empower operators to adjust UAS tasking based on real-time AI-assessed threats.
- **Red Teaming:** Repetitive OPFOR engagements testing survivability, autonomy, and adaptive decision-making.

Endstate

Constant repetitions at AVCCC provide feed mission logs and sensor data back into training datasets for REACT's model refinement. Each class could assign top operators to experimental teams testing new payloads, REACT modules, and swarm tactics. Graduates of this doctrine are more than drone operators, they are AI-enabled mission planners. As a part of the REACT-enabled force structure, they will:

- Integrate seamlessly into MDMP automation and ECOA analysis.
- Lead drone missions in multi-domain operations with minimal supervision.
- Serve as instructors and force multipliers across echelons.

Conclusions

REACT is under active development. Recent Department of Defense directives and contract awards have prioritized unmanned system procurement, operator training pipelines, and operational integration, thereby, positioning the REACT program to emerge as a central enabler in the Army's future force structure. We must adapt quickly as leaders to fully embrace current technology and leverage our people and institutions to prepare to fight and win our nations drone wars. The REACT initiative outlines the tool (material) to provide ground tactical commanders semi-automated military decision making process (MDMP) for timely and efficient drone offensive and defensive operations, the team (personnel) of Soldiers required at the Brigade echelon trained and proficient in drone warfare, and the training required to ensure that team's success in AI enhanced combined arms operations.[8]

Notes

[1]Jen Judson, "Anduril Wins 100M Deal to Build US Army's Next-Gen C2 Ecosystem," Accessed July 25, 2025, https://www.defensenews.com/land/2025/07/21/anduril-wins-100m-deal-to-build-us-armys-next-gen-c2-ecosystem/

[2]Anduril Industries, "Lattice Mesh," Accessed July 25, 2025, 2025, https://www.anduril.com/lattice-mesh/

[3]U.S. Department of Defense, *Unleashing U.S. Military Drone Dominance*, Memorandum from the Secretary of Defense to Senior Pentagon Leadership, Combatant Commanders, and Directors of Defense Agencies, Corrected copy. Controlled Unclassified Information. U.S. Department of Defense, Washington, DC, July 10, 2025

[4]Laura Gozzi, "How Ukraine carried out daring 'Spiderweb' attack on Russian bombers," Accessed July 29, 2025, https://www.bbc.com/news/articles/cq69qnvj6nlo

[5]RAND Corporation, *Hybrid Qualitative Simulation of Military Operations*, technical report, Accessed July 25, 2025 (Santa Monica, CA: RAND Corporation, 2024)

[6]RAND Corporation, *Modeling for Campaign Analysis: Lessons for the Next Generation of Models—Executive Summary*, technical report, Accessed July 25, 2025 (Santa Monica, CA: RAND Corporation, 2024)

[7]CPT Rachel Martin, "DCG Brief 24 JUL 25" (PowerPoint presentation, accessed July 25, 2025, 2025)

[8]Adams, Austin. 25 July 2025. Peer reviewed the paper to provide feedback on grammatical errors.

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